

RESEARCH ARTICLE

Plasma proteomic analysis of intermuscular fat links muscle integrity with processing speed in older adults

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Abstract

INTRODUCTION: More intermuscular fat (IMF) has been associated with lower cognitive performance through mechanisms that are not fully elucidated.

METHODS: The associations of 7628 plasma proteins with IMF were assessed in the Baltimore Longitudinal Study on Aging ($n = 941$, mean age = 66.7 ± 15.2) and validated in the Coronary Artery Risk Development in Young Adults Study ($n = 2451$, mean age = 50.2 ± 3.6). Processing speed was assessed by Digit Symbol Substitution Test (DSST). Associations between the main exposures, outcome, and mediators were evaluated using linear regression, and mediating effects were assessed by causal mediation analysis.

RESULTS: There were 722 plasma proteins associated with IMF in both the discovery and replication cohorts (false discovery rate [FDR] adjusted $p \leq 0.05$). Of these, 26 mediated the relationship between IMF and DSST, with effects ranging from 2.8% to 20.9% ($p \leq 0.05$). These proteins represented synaptic function and organization, and growth factor binding (FDR adjusted $p \leq 0.05$).

DISCUSSION: Reducing IMF may improve processing speed through effects on growth factor and synaptic activity.

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KEYWORDS

aging, epidemiology, intermuscular fat, processing speed, proteomics

Highlights

- Higher intermuscular fat is associated with lower processing speed, consistently across populations that represent different demographic characteristics.
- There is a robust plasma proteomic profile of intermuscular fat that is assessed in the abdominal and thigh skeletal muscle.
- The proteins associated with intermuscular fat reflect signals that are reflective of synaptic function and organization, as well as other molecular pathways such as growth factor binding.
- Circulating proteins partially explain the association between higher intermuscular fat and lower processing speed, suggesting that higher intermuscular fat effects processing speed through pathways including synaptic functions and growth factor binding.

1 | BACKGROUND

Aging is associated with decline in cognitive function across different domains.¹ Slowing rates of decline in cognitive function is an important public health goal, as cognitive decline is associated with unfavorable aging outcomes such as disability and dementia that could influence one's quality of life.² There is growing evidence linking skeletal muscle health with cognitive decline; thus, understanding the biology linking muscle with cognitive function may identify novel targets to slow the rates of cognitive decline with aging.^{3–5}

Intermuscular fat (IMF) increases with older age, and it has been associated with worse cognition in older adults, with the strongest evidence pointing to an association with slower processing speed.^{3–11} It has been hypothesized that the neurodegenerative effects of IMF may be triggered by inflammatory and cardiometabolic disorders. When adipocytes reach their fat storage capacity, greater adipose tissue accumulates ectopically, that is, in organs that usually have little adiposity, such as the skeletal muscle. As adipose tissue accumulates in skeletal muscle, both adipocytes and myocytes may release pro-inflammatory cytokines and these in turn can increase risk of systemic metabolic abnormalities such as insulin resistance and type 2 diabetes.^{12,13} In addition, it has been shown that myokines and adipokines (cytokines released from myocytes and adipocytes, respectively) have direct signaling effects on the central nervous system, influencing amyloid clearance and mitochondrial function in the brain.^{14–19} However, evidence is primarily from animal models or human studies at the extremes of muscle health (e.g., athletes or patient populations); the molecules linking IMF with poorer cognitive function have not been measured directly in community-dwelling older adults without overt disease.

To address this gap in knowledge, we aimed to characterize the plasma proteomic profile of IMF and explore whether IMF-associated

proteins mediate the relationship between IMF and cognitive function, focusing on processing speed. By taking an agnostic approach to explore the molecular pathways that are represented by the IMF-associated proteins, the results of this study provide mechanistic insights into the link between IMF and cognitive function.

2 | METHODS

2.1 | Study design and cohort description

The discovery cohort, the Baltimore Longitudinal Study of Aging (BLSA), is a rolling enrollment population-based study of participants residing predominantly in the Baltimore–Washington DC area.²⁰ Briefly, eligibility criteria include the absence of major chronic diseases (except for controlled hypertension) and cognitive or functional disabilities. Participants are followed continually at varying time intervals based on their age, namely every 4 years for participants younger than 60 years, every 2 years for participants between 60 and 79 years, and every year for participants older than 80 years of age or older. Demographic characteristics including age, sex, self-reported race, and years of education were obtained during a structured interview by BLSA staff. The study protocol (Protocol number 03-AG-0325) was approved by the National Institutes of Health Intramural Research Program Institutional Review Board and written informed consent was obtained from participants at each visit.

The validation cohort, coronary artery risk development in young adults (CARDIA) study began in 1985 with the recruitment of 5115 participants 18–30 years of age at field centers located in Birmingham, AL; Chicago, IL; Minneapolis, MN; and Oakland, CA.²¹ Recruitment was balanced for equal inclusion of Black and White as well as female and male participants, age (18–24, 25–30 years), and education

(≤ 12 years, > 12 years). The current study includes data from 3172 participants who agreed and received abdominal computed tomography (CT) scans at the year 25 (Y25) examination; of these, a total of 2390 had Digit Symbol Substitution Test (DSST), proteomics, and other key factors such as age, sex, race, and education. All participants provided written informed consent, and institutional review boards from each field center and the coordinating center approved the study annually.

2.2 | Assessment of intermuscular fat

Intermuscular fat (IMF) was quantified using CT scan of the thigh (BLSA) or abdomen (CARDIA). In the BLSA, imaging was performed by the radiology department of MedStar Harbor Hospital (Baltimore, MD) using a Somatom Sensation 10CT scanner (Siemens, Malvern, PA). A two-stage, deep-learning pipeline was implemented to estimate IMF from two-dimensional (2D) thigh CT slices.^{22,23} From this algorithm, the total muscle area and IMF was calculated as the average of the right and left thigh. In the CARDIA study, scans were performed using 64-channel multidetector GE CT scanners (GE Healthcare Milwaukee, WI) at the Birmingham, AL and Oakland, CA centers and Siemens CT scanners (Siemens, Erlangen, Germany) at the Chicago, IL and Minneapolis, MN centers. The protocol has been published previously.^{24,25}

2.3 | Cognitive function assessment

In both cohorts, the DSST was used to evaluate processing speed.²⁶ During the test, participants are initially shown nine pairs of digits and symbols. In the next sequence, participants are provided a series of digits and asked to draw the matching symbols for each digit. The score is the number of correct digit–symbol pairs identified during a period of 90 s. The DSST values were analyzed as a z-score.

2.4 | Proteomic assessment

Proteomic assessment in fasting plasma samples were conducted using the 7k SomaScan assay v4.1 (SomaLogic, Inc.; Boulder, CO, USA) in both the BLSA and CARDIA study, following a protocol that was detailed elsewhere.^{27,28} The final normalized dataset (hybridization control, median signal, plate-scale, inter-plate normalization) was used for the analysis. The final protein count consisted of 7268 target annotated human Slow Off-rate Modified Aptamers (SOMAmers) that were assessed and passed quality control in both the BLSA and CARDIA study. Some proteins were targeted by multiple SOMAmers; thus 6583 unique human proteins are represented in the panel. From this point forward, each SOMAmer will be referred as proteins in the article. Each protein relative abundance (Relative Fluorescence Units) was log2 transformed; outliers greater than 5 standard deviation (SD) from the mean were winsorized. The transformed and winsorized variables were scaled before analysis.

RESEARCH IN CONTEXT

- 1. Systematic review:** Many studies have shown that greater fat infiltration in the muscle, or intermuscular fat (IMF), correlates with decreased cognitive performance, especially in processing speed. The mechanisms that link IMF and cognitive function are not fully understood. In this study, we utilized plasma proteomic data to analyze the proteomic profile associated with IMF, aiming to identify the molecular pathways that connect IMF with cognitive function.
- 2. Interpretation:** The proteomic profile of IMF was highly reproducible across different studies and at different body sites. Enrichment analyses of IMF-associated proteins show overrepresentation of proteins involved in synaptic integrity and growth factor binding pathways, suggesting that these processes may underly the poor cognitive performance in individuals who have higher IMF.
- 3. Future directions:** Our cross-sectional analysis indicates that the IMF could potentially impact processing speed by affecting synaptic function and growth factor binding pathways. Future research should utilize longitudinal data to delve deeper into the timing of these relationships and investigate whether these proteins correlate with cognitive function trajectories.

2.5 | Statistical analysis

All analyses were conducted using R version 4.2.2. The associations between IMF and plasma proteins or DSST were assessed using linear regression models. For proteomic analysis of IMF, the winsorized protein value was the dependent variable and IMF was the independent variable in the regression model. The models were adjusted for age, sex, muscle area (or muscle volume for CARDIA), self-reported race, and years of education. To understand whether the relationship between IMF and DSST is modified by self-reported race or sex, we included an interaction term between IMF*race or IMF*sex in the regression model. For the proteomic analysis, the p-value was adjusted using false discovery rate (FDR), and an FDR adjusted $p \leq 0.05$ was considered statistically significant. Proteins that were significantly associated with IMF in the BLSA were tested for validation in CARDIA study. Significant validation was considered at FDR adjusted $p \leq 0.05$. To test whether the IMF-associated proteins mediate the association between IMF and DSST, a causal mediation analysis was conducted using R regmedint package (version 1.0.1).²⁹ For each model, IMF was the main exposure variable, DSST the dependent variable, and individual protein was the mediator. Mediation was considered statistically significant when a significant indirect association for IMF on DSST was observed ($p \leq 0.05$).

TABLE 1 Characteristics of participants from the discovery (Baltimore Longitudinal Study on Aging) and validation (Coronary Artery Risk Development in Young Adults [CARDIA]) studies.

	BLSA	CARDIA
N	941	2445
Age (years)	66.7 (15.2)	50.2 (3.6)
Sex (%Male)	46.9% (441)	43.2% (1057)
BMI (kg/m ²)	27.0 (4.6)	30.1 (7.0)
Intermuscular fat (cm ²)	20.1 (9.5)	18.4 (12.6) ^b
Muscle area (cm ²)	220.9 (62.5)	143.6 (37.1) ^c
Digit Symbol Substitution Test	45.9 (13.5)	68.1 (16.4)
Self-reported race (%)		
White	67.1% (631)	54.7% (1338)
Black	24.9% (234)	45.3% (1107)
Other ^a	8.1% (76)	NA
Years of education	17.7 (2.7)	15.1 (2.7)

Note: Mean (SD) or *n* (%).

^aOther includes participants who identified as American Indian or Alaska Native, Chinese, Filipino, Hawaiian, Japanese, Other Asian or Other Pacific Islander, Other non-White, or Not Classifiable.

^bIntermuscular fat volume (cm³).

^cMuscle volume (cm³).

Over-representation analysis (ORA) was conducted to determine whether any pathways were enriched among the IMF-associated proteins. Using the proteins measured by the SomaScan platform as background, we used the 716 IMF-associated proteins to explore enrichment of Gene Ontology (GO) biological processes (BPs), cellular component (CC), and molecular function (MF) libraries. Enrichment was considered statistically significant at FDR adjusted $p \leq 0.05$. The ORA was performed using R ClusterProfiler package (version 4.6.2).

3 | RESULTS

3.1 | Association between intermuscular fat and DSST

The average age of the discovery cohort was older than the validation cohort (Table 1; 66.7 ± 15.2 vs 50.2 ± 3.6). The age difference is reflected in processing speed, with higher DSST scores in the CARDIA study compared to BLSA. Despite these differences, IMF was found to be statistically significant in both studies. In the BLSA, there was significant negative relationship between IMF and DSST (β , 95% CI = -0.11 , -0.15 to -0.05), indicating that higher IMF was associated with slower DSST. Similarly, in the CARDIA study a negative relationship was also observed (β , 95% CI = -0.09 , -0.12 to -0.05). The association between IMF and DSST was not modified by sex ($P_{\text{IMF} \times \text{SEX}} = 0.731$) or self-reported race ($P_{\text{IMF} \times \text{Race}} = 0.560$).

3.2 | Proteomics of intermuscular fat

In the discovery cohort, of 7268 SOMAmers (6375 proteins) assessed, 782 SOMAmers (710 proteins) were significantly associated with IMF (Table S1, Figure 1A). There were 321 SOMAmers (297 proteins) found in higher abundance with greater IMF, and 461 SOMAmers (413 proteins) in lower abundance with higher IMF. These 782 SOMAmers were tested for association with IMF in the CARDIA study, and 722 (654 proteins; 92.3%) were found to be significantly associated. For all proteins, the direction of association was consistent, with high correlation ($r = 0.95$, Figure 1B) between the beta estimates from the regression models from the two studies for the 782 SOMAmers that were tested for replication.

An ORA based on GO terms for the 654 proteins represented by the 722 IMF-associated SOMAmers identified multiple pathways (Figure 1C). The GO- BP terms encompass two broad categories of pathways that represent synaptic biology ("regulation of synapse assembly," "regulation of synapse structure and activity," and "axon development") and inflammation ("taxis" and "chemotaxis"). The GO-CC terms complement the results from synaptic function from the BP that include CC terms such as "intrinsic component of synaptic membrane, γ -aminobutyric acid (GABA)-ergic synapse" and "axonal growth cone," as well as indicator of extracellular matrix. Finally, the GO-MF identified included "glycosaminoglycan binding" and "growth factor binding," and terms representing general signal transduction activities.

3.3 | Proteomic mediators of intermuscular fat and DSST

Among the 722 IMF-associated SOMAmers, 80 and 370 proteins were associated with DSST in BLSA and the CARDIA study, respectively, with 57 proteins significantly associated in both studies (Table S2, Figure 2). For the 57 common proteins, the direction of associations was consistent across studies, with 37 proteins associated with faster processing speed and 20 with slower processing speed. Furthermore, the 37 proteins associated with faster processing speed were associated with lower IMF; conversely the 20 proteins associated with slower processing speed were associated with higher IMF (Figure 2A,B). An ORA of the 57 proteins shows enrichment of pathways similar with the IMF-associated proteins including GO-BP terms "chemotaxis," and "regulation of presynaptic organization/assembly"; GO-CC terms "intrinsic component of postsynaptic membrane" and "receptor complex"; and, finally, GO-MF terms "growth factor binding" and "signaling receptor activity" (Figure 2C).

Of the 722 IMF-associated SOMAmers, 26 SOMAmers representing 24 proteins significantly mediated the relationship between IMF and DSST in both BLSA and the CARDIA study (Table S3, Figure 3). The proportion mediation ranges were 7.2%–20.9% in BLSA and 2.8%–8.9% in the CARDIA study. For all but one protein, the directions of indirect effects were negative in both the BLSA and CARDIA studies. Among the 24 mediating proteins, 14 were found in lower abundance with higher IMF, and associated with higher DSST. Conversely, 10 pro-

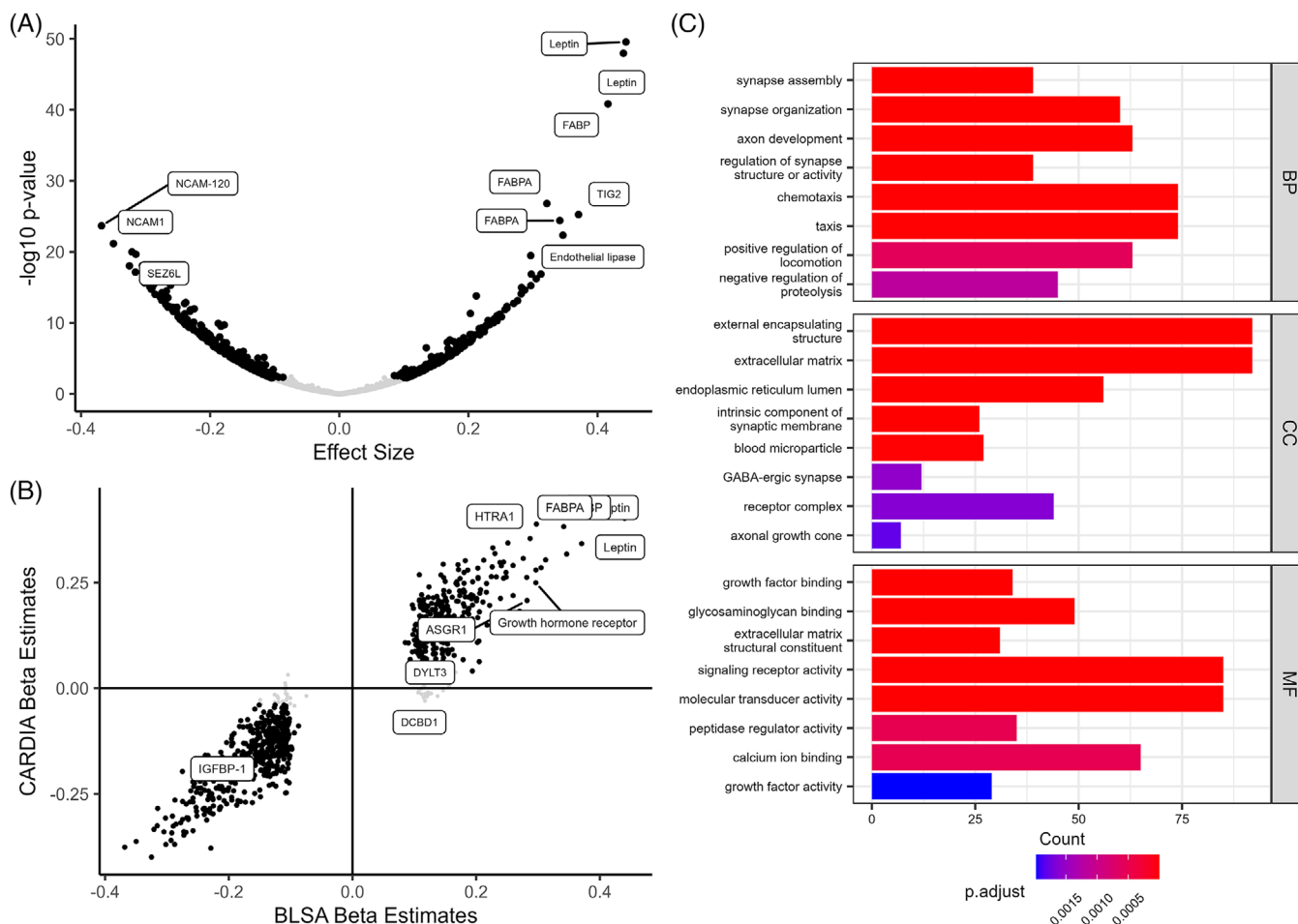


FIGURE 1 Plasma proteomic profile of intermuscular fat (IMF). Associations between IMF and plasma proteins were tested in participants of the BLSA ($n = 941$) and validated in the CARDIA study ($n = 2445$). (A) In the discovery BLSA, of the 7268 SOMAers assessed, 782 were associated with IMF. (B) Of the 782 SOMAers, 722 were validated in the CARDIA study. The associations between the beta estimates from the regression between IMF and SOMAers in the BLSA and the CARDIA study was high ($r = 0.96$). (C) Enrichment analysis of Gene Ontology (GO) biological process (BP), cellular component (CC), and molecular function (MF) of the 722 SOMAers (representing 654 unique proteins) show enrichment in neurological pathways and growth factor binding. BLSA, Baltimore Longitudinal Study of Aging; CARDIA, Coronary Artery Risk Development in Young Adults; SOMAers, Slow Off-rate Modified Aptamers.

teins were found in higher abundance with greater IMF and associated with lower DSST. AnORA of the 24 mediating proteins showed no significantly enriched pathways.

4 | DISCUSSION

In this study, we explored the plasma proteomic profiles of IMF, which showed that there is a distinct proteomic signature of 654 proteins in two independent cohorts of older adults living in the community. The protein signature was highly consistent despite IMF being measured in skeletal muscle at different locations. A subset of 24 IMF-associated proteins explain, at least in part, the association between higher IMF and slower processing speed. Taking an agnostic approach, enrichment analyses of the proteins identified indicate that pathways such as synaptic function and growth factor activity may be the link between IMF and processing speed.

The study results confirm the growing evidence supporting the associations between higher IMF and lower cognitive function, particularly with slower processing speed. An initial study reported inverse associations between measures of IMF using peripheral quantitative CT (pQCT) with psychomotor function, visual learning, and overall cognitive function.³⁰ Subsequent studies have shown similar associations in different populations, particularly with processing speed and with other indicators of brain health such as neuroimaging markers.^{31,32} Our study used proteomic data to explore the mechanisms that may link higher IMF with slower processing speed.

One intriguing insight from our analysis was that the proteome associated with IMF was enriched with proteins involved in synaptic organization and function. The four proteins (SLITRK3, NOTCH1, C1QL3, and EPHA4) that significantly mediated the association were found in lower abundance with higher IMF and lower processing speed. These proteins are widely expressed across the central nervous system, with some also found in muscle tissue.^{33–36} They play crucial roles

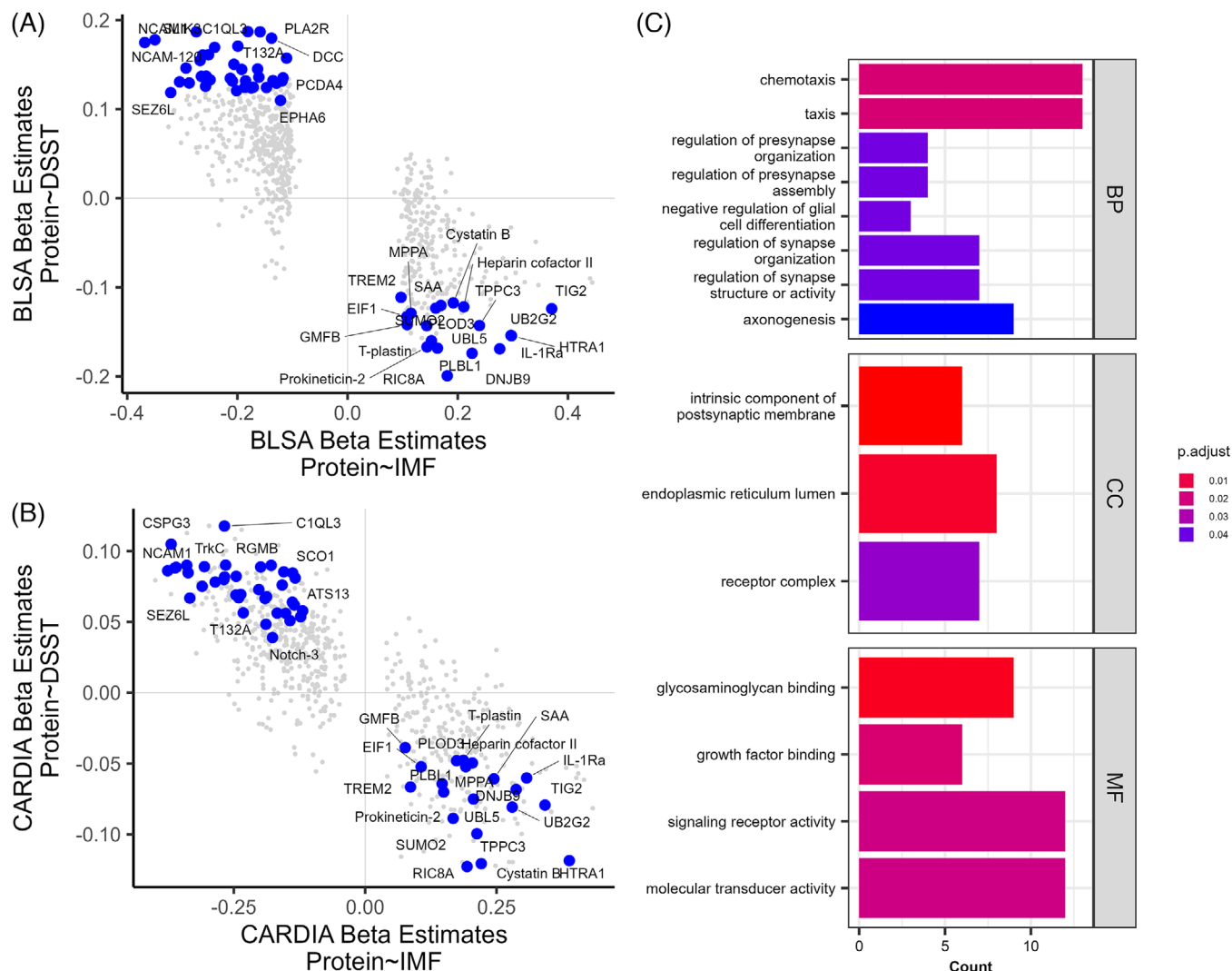


FIGURE 2 Plasma proteomic profile of DSST. Associations between intermuscular fat (IMF)-associated plasma proteins and DSST were tested in participants of the BLSA ($n = 941$) and CARDIA study ($n = 2445$). There were 57 proteins (blue circles) associated with DSST in the BLSA (A) and CARDIA (B) study. In both cohorts, 37 proteins were associated with faster DSST and lower IMF, and 20 proteins associated with slower DSST and higher IMF. (C) Enrichment analysis of GO biological process (BP), cellular component (CC), and molecular function (MF) of the 57 SOMAers show consistent pathways to the analysis of IMF-related proteins. BLSA, Baltimore Longitudinal Study of Aging; CARDIA, Coronary Artery Risk Development in Young Adults; DSST, Digit Symbol Substitution Test.

in brain function by influencing various processes: SLITRK3 regulates neurite outgrowth³⁴; C1QL3 participates in neuronal organization as trans-synaptic cell adhesion complexes³⁵; NOTCH is involved in neuronal stem cell maintenance and activation^{33,36}; and EPHA4 is crucial for synaptic function and integrity through regulation of bidirectional communication between neurons and astrocytes.³⁷ SLITRK3 knock-out mice have impaired postsynaptic neurotransmission, resulting in multiple abnormal hypermobile movements.³⁸ C1QL3 knockout animals display fewer excitatory neurons compared to wildtype, and they exhibited stunted fear conditioning as well as a modest deficit in motor learning.³⁹ In humans, several lines of evidence support the link between these proteins with cognitive function. First, a lower abundance of soluble NOTCH1 protein was reported in patients with AD; further protein abundances were positively correlated with Mini-

Mental Status Examinations (MMSE).⁴⁰ A mendelian randomization study supported the causal association between SLITRK3 and general cognitive function.⁴¹ Taken together, our data suggest that lower levels of proteins involved in synaptic function and integrity simultaneously manifest as lower integrity in the skeletal muscle (higher IMF) and central nervous system (slower processing speed). Our results of causal modeling analyses indicate a potential direction of these associations, whereby higher IMF may affect processing speed by lowering the circulating levels of neuroprotective proteins.

Among the proteins that were found in higher abundance in relation to higher IMF and lower DSST (e.g., with potential neurodegenerative effects), the myokine high-temperature requirement A1 (HTRA1) is notable for its proteolytic effects on extracellular matrix remodeling skeletal muscle, brain, and adipose tissue.^{42,43} HTRA1 also has

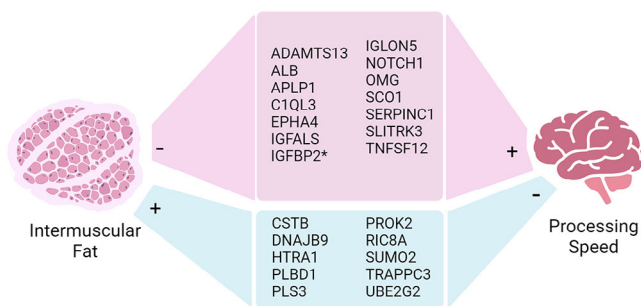


FIGURE 3 Association between intermuscular fat (IMF) and processing speed mediation by plasma proteins. The association between degree of IMF and processing speed assessed using DSST was significantly mediated by 24 proteins in both the BLSA and CARDIA study. Except for IGFBP-2, all proteins show negative indirect effects. Red: There were 14 proteins that had negative associations with IMF (lower levels for higher IMF) and positive association with DSST, indicating a potential neuroprotective role. *IGFBP-2 was the only protein with a negative association with DSST in the CARDIA study. Blue: There were 10 proteins with positive associations with IMF (higher levels for higher IMF) and negative associations with DSST, indicating a potential neurodegenerative role. Created with BioRender.com. CARDIA, Coronary Artery Risk Development in Young Adults; DSST, Digit Symbol Substitution Test.

inhibitory effects on the transforming growth factor- β superfamily. In a translational proteomic study of several thousand proteins,⁴⁴ HTRA1 was correlated positively with brain amyloid. Several HTRA1 mutations have been associated with cerebral small vessel disease in humans,⁴⁵ but these studies did not test associations with cognitive function. We and others have shown that higher cerebral small vessel disease is strongly associated with lower processing speed; thus, it is possible that higher IMF leads to higher levels of the myokine HTRA1; this in turn drives cerebral small vessel disease, manifesting as slower processing speed.

The finding that proteins related to insulin-like growth factor (IGF) are related to IMF and cognition has been shown previously. Of note, IGF-binding protein 2 (IGFBP-2) was positively associated with skeletal muscle mass and strength in prior proteomic studies in South Africa⁴⁶ and Germany.⁴⁷ Conversely, its role in relation with cognition is less clear. In a recent study of older adults,⁴⁸ IGFBP-2 was associated with smaller hippocampal volume among amyloid-negative adults, but not amyloid-positive adults, indicating that the influence on IGFBP-2 on hippocampal volume varies depending on severity of brain pathology. Conversely, others found a positive association with tau.⁴⁸ The association of IGFBP-2 with cognitive status also varies across studies, presumably because the role of IGFBP-2 varies depending on developmental stages.^{49,50} Differences in age and/or in brain pathology may explain why in our study IGFBP-2 had an inverse association with DSST in the CARDIA study (younger) but not in BLSA (older).

Prior studies have linked IMF with factors that influence cardiometabolic health, in particular serum cytokine levels and adipokines.¹³ Both muscle and the fat that infiltrate muscle have secretory capabilities including pro- and anti-inflammatory cytokines that have been associated with cognitive health.⁵¹ Moreover, cytokines

released by skeletal muscles and adipocytes with neuronal signaling properties can cross the blood-brain barrier.⁵² Consequently, some studies explored these proteins as possible mechanism linking IMF with cognitive function. In the Geelong Osteoporosis Study (GOS), tumor necrosis factor α (TNF α) explained the association between muscle density measured in the radius and psychomotor function, but the mediation was not established for the muscle density measured in the tibia.³⁰ In addition, other inflammatory markers, interleukin (IL)-6, IL-10, and IL-13 did not explain the association in the GOS. In other studies, although no formal mediation analysis was conducted, the association between IMF and cognitive function was robust to adjustment for cardiometabolic conditions, including hypertension, diabetes, obesity, and measures of inflammation (leptin, adiponectin, and IL-6).⁴ In the current report, IL-6 was moderately associated with IMF in the BLSA, and C-reactive protein (CRP) was negatively associated with IMF in both the BLSA and CARDIA study, but they were not significant mediators. It is intriguing that enrichment analysis did not show inflammation as a major mediating pathway linking IMF with processing speed.

Although our findings provide strong biological evidence linking IMF with processing speed, there are several important limitations. First, this is a cross-sectional study, thus we cannot make causal inferences on the associations we observe. For example, it is possible that higher IMF may drive lower levels of proteins, which in turn compromise synaptic integrity in the brain, ultimately manifesting as slower processing speed. Alternatively, there is another factor that alters homeostasis of proteins engaged in synapsis remodeling among older adults with higher IMF. Furthermore, there may be a bi-directional relationship—individuals with lower cognitive function and processing speed may have lowered synaptic activity leading to decreased muscle quality. In addition, the protein that mediates IMF with DSST may differ throughout the life course. Future studies should examine these relationships using longitudinal data. Although our study uses direct measurements of IMF using CT scans, we cannot exclude the possibility that the correlations we observed are not driven by fat infiltrations in the muscle per se, but through overall adiposity. The analysis was not adjusted for body mass index (BMI), as total adiposity may be part of the casual pathways linking IMF with processing speed; thus inclusion of BMI in the regression model would introduce an overadjustment bias.⁵³ Our study has several strengths. First, our results were consistent across two studies that have different demographics from the standpoint of age, self-reported race, and geographical distributions. In both studies, IMF was measured using CT scan, providing us with an objective measure. Finally, although the proteomic assessment did not measure all circulating proteins, the analysis targeted large numbers of proteins using the same assessment method. Future studies should explore validating these findings in longitudinal studies to determine the temporal relationship between protein abundance, IMF, and processing speed.

By leveraging data on circulating proteins, we provide strong evidence suggesting that IMF affects processing speed through inflammation; regulating of growth factor activity; and synaptic organization, function, and activity. These data support the interplay between mus-

cular quality and the brain and suggest that the muscle is an important target for prevention of cognitive decline associated with aging.

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CONFLICT OF INTEREST STATEMENT

The authors have no conflicts to declare. Author disclosures are available in the [Supporting Information](#).

CONSENT STATEMENT

All participants of the study in this report provided written informed consent.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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